Fiscal Year:	FY 2022	**Task Last Updated:**	FY 03/01/2023
PI Name:	Contractor, Noshir Ph.D.		
Project Title:	CREWS: Crew Recommender for Effective Work in Space		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	217-390-6270
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City:	Evanston	**State:**	IL
Zip Code:	60208-0834	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	07/01/2015	**End Date:**	09/30/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2022 per A. Beitman/HFBP and NSSC information (Ed., 10/20/21) NOTE: End date changed to 9/30/2021 per NSSC (Ed., 4/1/21) NOTE: End date changed to 3/31/2021 per NSSC (Ed., 5/21/2020) NOTE: End date changed to 6/30/2020 per NSSC (Ed., 10/10/19) NOTE: End date shows 6/30/2019 in NSSC (Ed., 4/2/19) NOTE: End date changed to 5/17/2019 per D. Arias/HRP (Ed., 3/22/18) NOTE: Element change to Human Factors & Behavioral Performance; previously Behavioral Health & Performance (Ed., 1/17/17)		
Key Personnel Changes/Previous PI:			

COI Name (Institution):	Bell, Suzanne Ph.D. (DePaul University) DeChurch, Leslie Ph.D. (Northwestern University)
Grant/Contract No.:	NNX15AM32G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Team composition, the configuration of member attributes and their relationships, is a critical enabling feature of fostering effective teamwork and likely to play an important role in the effectiveness of future long-duration space exploration (LDSE). Limited research on team composition in environments analogous to LDSE exists, and currently how team composition can be used to optimize crew functioning and performance is unclear. Our research aims to: (1) identify the effects of team composition on team functioning in LDSE and the critical factors of team composition driving this effect, (2) identify particular patterns of this effect with different team compositions, (3) identify methods for composing teams for LDSE, (4) develop a predictive team composition model for use in composing teams and identify potential issues with already composed teams, and (5) provide recommendations for composing teams for LDSE. To address these critical aims, we propose a 3-year, multi-method research effort, in which we: (1) develop an agent-based model of team composition for LDSE based on empirical data linking key model inputs (e.g., individual difference variables, network relational factors, task characteristics) to team functioning (e.g., social integration, team processes, team cohesion, team conflict) in LDSE-relevant contexts; (2) conduct virtual experiments using characteristics and relationships identified in Phase I to identify the team functioning patterns that arise under different member compositions, and create a predictive model of team composition; and (3) conduct an initial validation of the model developed in Phase 2 in the Human Exploration Research Analog (HERA) and NASA Extreme Environment Mission Operations (NEEMO) analogue environments using specific manipulations of key factors (e.g., compositions; situational characteristics). Research products critical to closing Team Gap 8 will be developed including a predictive model of team composition in LDSE, evidence in support of the model, and a mockup of a proposed interface to assist in the staffing and management of LDSE crew and mission teams.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	While the primary objectives of this project are to be applied to astronaut crews in LDSE contexts, results from this research may also benefit teams on Earth in similar ICE (Isolated, Confined, and Extreme) conditions. Teams such as those sent to winter-overs in Antarctica or submarine crews that spend months underwater would be analogous environments in which the results of this research may prove useful. In a general sense, our findings could have implications for composing optimal teams that are not in ICE conditions, such as work teams at an organization, teams of students working on a project, teams of scientists, and squadrons of military personnel, to give but a few examples.
Task Progress:	Executive Summary of Phase 1: Through our Phase 1 efforts, we make one contribution to team risk in long-distance space exploration missions. We developed a computational model that successfully predicts crew social relations based on personal, task, social, and situational factors. The agent-based computational model (ABM) simulates social relationships in crews as they complete their work, as well as the evolution of these social relationships during long-duration space exploration (LDSE). The model was calibrated using data gathered from crew completing missions in LDSE analogs, on Earth, over a period of 30 to 45 days (Deliverable 1). Executive Summary of Phase 2: Through our Phase 2 efforts, we make one contribution to team risk in long-distance space exploration missions. We finalized the development of the computational model and published the report detailing the methodology and results of computational modeling (Deliverable 2). Executive Summary of Phase 3: Through our Phase 3 efforts, we make seven contributions to team risk in long-distance space exploration missions. First, we used the CREWS ABM developed and demonstrated how empirically driven ABMs enable the recommendation of countermeasures that can support teams, in this case, to promote positive and dissuade negative social relationships within teams. We considered a “re-pairing” countermeasure, in which we reassigned which pairs of crew members are assigned to collaborate based on simulation results from the ABM. We tested the effectiveness of these ABM-recommended countermeasures among four member crews completing 45-day missions in the HERA LDSE analog (Deliverable 3). Our second contribution was to demonstrate the validation, reliability, and impact of the predictive team model as well as practical and research recommendations (Deliverable 4). The third contribution is to examine the crew performance over time (Deliverable 4). We found that behavioral team performance (cognitive conflict and psychomotor tasks) increases over time, whereas conceptual team performance (creativity and intellective tasks) declines. The fourth and fifth contributions are to examine which crew network patterns predict team performance (Deliverable 5 & Deliverable 6). One of the findings was that densely configured positive working networks (i.e., “Who do you enjoy working with?”) are associated with high performance on tasks, such as executing tasks (i.e., simulation tasks in which two crew members use a joystick to fly a transit vehicle to collection sites, while the other two crew members use virtual reality goggles to complete Extra Vehicular Activity exploring an asteroid’s surface). The sixth contribution is to examine the effect of team size on the performance, motivation, and effort of individual teammates (Deliverable 7). We found that there are many considerations for crew size, but solely from a team perspective, there may be some advantages to smaller teams (e.g., more influenced by teammates' ideas, greater contributions to performance, less social loafing, greater viability). The seventh contribution is that we developed a prototype dashboard for the detection of crew social relations, available here: http://nasa.soc.northwestern.edu/ (Deliverable 8).
Bibliography Type:	Description: (Last Updated: 03/17/2026)
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